NOTES:

1. ALTHOUGH CIR IS DESIGNED TO THE MDP OF 2015 PSIA ON THIS PATH, THE PRESENCE OF RVA1 AS ACME INSTALLED HARDWARE CHANGES THE MDP OF THIS PATH TO 135 PSIA.

MDP REGIONS	
A	2015 PSIA
B	1815 PSIA
C	200 PSIA
D	135 PSIA
E	115 PSIA
F	100 PSIA
G	RESERVED
H	40 PSIA
I	30 PSIA
J	15 PSIA

○ – START/END MDP REGION

/ – MDP REGION BOUNDARY

MDP REGIONS ARE DEFINED PER CIR FLIGHT SAFETY DATA PACKAGE CIR-DOC-3850. 2 FAULT TOLERANCE REQUIRED FOR BOUNDARY TRANSITION.

GN2 FROM ISS GDS

PRESSURE: 517 TO 827 kPa (75 TO 120 psia)

FLOW: 5.43 kg/hr (12lbm/hr), 90.16 SLPM

TEMPERATURE: 15.6°C TO 45°C (60°F TO 113°F)

UTILITY INTERFACE PANEL (UIP) — RACK UTILITY PANEL (RUP) — GAS INTERFACE PANEL (GIP)

TO ISS VES OVERBOARD VENT — QD24 — FH4 ARIS VACUUM EXHAUST UMBILICAL 683-61634-2 — QD23 — D PR9 H 36 PSIA VES REGULATOR ASY 67211MFA030061 — QD22 — FH1 CIR VES HOSE 67211MFA030007 — QD5

MANUAL VENT VALVE MANIFOLD 67212MFAM22000: OR1, Cv=1.89, MV4 1/2" 3-WAY BALL VALVE

VENT MANIFOLD 67212MFAM20000: 136 kPa (19.7 PSIA) INCREASING, PT31, PS7, SV29, 100 slpm MFC5, SV34, DPS1, TM11, QD21

MANUAL VENT VALVE MANIFOLD 67212MFAM22000: PT13, QD6, MV10, MV9, QD7

ADSORBER CARTRIDGE 67212MFAM71000

EXHAUST MANIFOLD 67212MFAM21000: PT30, SV24, SV11, SV22, CV9, SV23

EXHAUST MANIFOLD 67212MFAM21000: SV1, CV12

FOMA CAL UNIT 67212MFAM62000: TM13, PT34

TM9 PUMP 1 CV14 67212MFAM23000; TM10 PUMP 2 CV15 67212MFAM23000; F7

GC INSTRUMENTATION PACKAGE 67212MFAM30000: PT14, CV4, CV3, PT21, 15 PSIA PR5, SV32, SV33, PT23, 207 kPa (30 PSIA) INCREASING, PS6, F9, SAMPLE IN, PUMP 4 VENT, PUMP 5 VENT, He IN, Ar IN, RE-PACKAGED COMMERCIAL GC SEE 67212MFSM10003, COLUMN VENTS, PR10 15 PSIA, SV30, SV31, PUMP 3, 20 PSIA RV7, PR11 15 PSIA

GC CHECK GAS SUPPLY MANIFOLD 67212MFAM33000: SV27, RV5, RV6, PT22, PR6 65 PSIA, PT24, MUF5, MV15, F10, QD8, MV8, GB6 CHECK GAS 12518 kPa (1815 PSIA) Max Pressure; QD18

GC HELIUM SUPPLY MANIFOLD 67212MFAM32000: SV25, RV2, RV1, PT20, PR8 95 PSIA, PT27, MUF6, MV16, F11, QD9, MV6, GB4 Helium 12518 kPa (1815 PSIA) Max Pressure; QD19

GC ARGON SUPPLY MANIFOLD 67212MFAM31000: SV26, RV3, RV4, PT19, PR7 95 PSIA, PT26, MUF7, MV17, F12, QD10, MV7, GB5 Argon 12518 kPa (1815 PSIA) Max Pressure; QD20

FUEL/PREMIXED FUEL MANIFOLD #4 67212MFAM10000: TM5, 793 kPa (115 PSIA) INCREASING, PS2, GB2 Fuel 13898 kPa (2015 PSIA) Max Pressure, MV2, QD2, F2, SV6, PR2 90 PSIA, SV7, PT6, 2.0 slpm MFC1 MASS FLOW CONTROLLER, SV8 TIMER CONTROLLED, MV14, MUF4, PT4, PT5, QD14

DILUENT/PREMIXED GAS MANIFOLD #1 67212MFAM12000: TM7, 793 kPa (115 PSIA) INCREASING, PS3, GB3 Diluent 13898 kPa (2015 PSIA) Max Pressure, MV3, QD3, F3, SV10, PR3 100 PSIA, SV9, PT9, 30 slpm MFC2 MASS FLOW CONTROLLER, SV17, CV6, MV13, MUF3, PT7, PT8, QD15

HIGH PERCENTAGE OXYGEN MANIFOLD #2 67212MFAM11000: TM6, 793 kPa (115 PSIA) INCREASING, PS1, GB1 Oxygen 13898 kPa (2015 PSIA) Max Pressure, MV1, QD1, F1, SV3, PR1 100 PSIA, SV4, PT3, 30 slpm MFC3 MASS FLOW CONTROLLER, SV5, CV7, MV12, MUF2, PT1, PT2, QD16

NITROGEN HIGH PRESSURE MANIFOLD #3 67212MFAM13000: TM8, 793 kPa (115 PSIA) INCREASING, PS4, GB7 NOT USED ONLY ISS GN2, GB7 Nitrogen 13898 kPa (2015 PSIA) Max Pressure, MV15, QD4, F4, SV13, PR4 100 PSIA, SV12, PT12, 30 slpm MFC4 MASS FLOW CONTROLLER, SV2, CV8, MV11, MUF1, PT10, PT11, SV28, QD17, QD28

STATIC MIXER PACKAGE 67212MFAM14000: STATIC MIXER, SV14

QD25 — UTILITY INTERFACE PANEL (UIP) — FH3 ARIS GN2 UMBILICAL 1F15193-1 — QD26 — RACK UTILITY PANEL (RUP) — GAS INTERFACE PANEL (GIP) — MV19 — FH2 CIR GN2 HOSE 67211MFA030005 — 517 TO 827 kPa (75 TO 120 PSIA) — QD27 — VENT TO ISPR

RVA1 135 PSIA

VALVE TIMER RESET — 13 kPa (1.9 PSIA) DECREASING — PS8, PT28, TM3 — PT29, PS9, TM4 — VALVE TIMER RESET

7AN CHAMBER: 105 LITERS MDP=931 kPa (135 psia)

ACME CHAMBER INSERT ASSEMBLY S1101MFA00100: PSA2 22 PSIA OPENS ON ASCENDING IGNITER INHIBIT; PTA3; PSA1 6 PSIA CLOSES ON ASCENDING IGNITER INHIBIT; PTA1 To Outer Coflow; PTA2 To Inner Coflow; To Gas-Jet Burner; To Spherical Burner

SV9 SV8 SV5 SV4 SV7 SV6 SV3 SV1 SV2 MFCA3 MFCA1 MFCA2 QD A9 QD A8 QD A7 QD A3 QD A2 QD A1 QD A6 QD A5 QD A4

SV15, F8, PS5 793 kPa (115 PSIA) INCREASING, PT116, TM2, QD11, QD13, QD12, TM1, PT117, INTERFACE RESOURCE RING

SV21, PT32, F5, QZS1 QZS, SPIDER MANIFOLD 67212MFAM10204, F6, MV5 MANUAL VENT 3 POSITION/3 WAY BALL VALVE

PT125, SV16, CV1; PT15, SV18, CV2; PT133, SV20, CV10

EXPERIMENT ASSEMBLY 67212MFAH00002

PIPING COMPONENT SYMBOLS			
BUTTERFLY VALVE	DIRECTIONAL VALVES 3-WAY 4-WAY	CYLINDER REGULATOR	PRESS GAUGE
CHECK VALVE			FILTER
GATE VALVE		TURB FLOWMETER	STRAINER
RELIEF VALVE		ORIFICE	HEAT EXCH
BALL VALVE	PRESSURE REGULATOR	VENTURI	PUMP
GLOBE VALVE		RUPTURE DISK	MOTOR
3-WAY VALVE	LOADER		

ACTUATOR SYMBOLS
CYLINDER
HAND
PNEU CONTROL
ROTARY MOTOR
SOLENOID

PRIMARY CONTROL ELEMENT SYMBOLS			
EP	ELEC-PNEU XDCR	PS	PRESSURE SWITCH
EH	ELEC-HYDR XDCR	PT	PRESSURE XDCR
LLS	LIQ LEVEL SWITCH	TC	THERMOCOUPLE
LLT	LIQ LEVEL XDCR	TS	TEMP SWITCH
LS	LIMIT SWITCH		
PDS	PRESS DIFF SWITCH		
PDT	PRESS DIFF XDCR		

ITEM DESIGNATIONS
LOCALLY MOUNTED
CONTROL ROOM MOUNTED
ALTN CONTROL ROOM MOUNTED
REF DRAWING

INSTRUMENT LETTER DESIGNATIONS FOR ○ ⬡ ◇			
FIRST LETTER	B – BURNER	L – LEVEL	Z – POSITION
	E – EMERGENCY	P – PRESSURE	J – POWER
	F – FLOW	S – SPEED	U – SURGE
	H – HAND	T – TEMPERATURE	
SUCCEEDING LETTER	A – ALARM	I – INDICATE	S – SHUTDOWN
	C – CONTROL	L – LOW	Y – DATALOG
	D – DIFFERENTIAL	P – PERMISSIVE	
	H – HIGH	R – RECORD	

INSTRUMENT SYMBOLS	
GENERAL	DATA COLLECTOR FACILITY
PRGM LOGIC CONTROLLER	PRGM PROCESS CONTROLLER

NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED: 1. ALL DIMENSIONS ARE IN INCHES 2. BREAK EDGES .010-.020 3. MAX SURFACE FINISH 63 MM 4. LINEAR TOLERANCES: .X ± 0.03; .XX ± 0.01; .XXX ± 0.005 5. ANGULAR TOLERANCE ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL; DO NOT REVISE WITHOUT AN APPROVED ECO.

APPROVED: R. TURK 10/3/14 — PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	M. HOMKO 9/18/14	QUALITY	D. SEELEY 9/17/14
REVIEWER	G. FEDOYCH 9/18/14	MATERIALS	M. WILKINSON 9/18/14
MECHANICAL	A. DRAKE 9/18/14	PROJ LEAD	B. BOROWSKI 9/29/14
ELECTRICAL	T. ZORELI 9/17/14		
STRUCTURAL	C. WERNER 9/24/14		
SAFETY	M. BORRELLI 9/17/14		

DIMENSIONS AND TOLERANCING PER ASME Y14.5M-1994

CONTRACT NUMBER: NNC14CA02C
PROJECT CODE: SZ-ISS

ZIN TECHNOLOGIES 6745 ENGLE ROAD AIRPORT EXECUTIVE PARK CLEVELAND, OHIO 44130 (440) 625-2200 www.zin-tech.com

REV	DESCRIPTION	APPROVED
A	REVISED PER ACME-ECO-376	MH 10/3/14, AD 10/3/14, KR 10/3/14

REVISION HISTORY

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE: ACME/CIR INTEGRATED PNEUMATIC SCHEMATIC

SIZE	CAGE CODE	DWG NO	REV
E	0SG68	S1101MFS0004	A

SCALE NONE | INITIAL REL DATE 09/29/14 | SHEET 1 OF 1

RELEASE LEVEL: RELEASED
MODEL NAME: S1101MFS0004_rev_A.DWG

AutoCAD 2012

REV A | SH 1 | DWG NO S1101MFS0004